Project Introduction

Electric Vehicles in Washington State

Abbreviations: EV = electric vehicle; BEV = battery electric vehicle; PHEV = plug-in hybrid electric vehicle; DOL = Washington State Department of Licensing.

1. Literature Review

Many studies ask why people buy electric vehicles (EVs). Most of them use regression models to test which factors are linked to adoption. Sierzchula et al. (2014) studied the 2012 EV market share in 30 countries. They found that financial incentives, charging stations and local car production were all linked to a higher market share, and that charging stations had the strongest link. They also found that incentives or charging alone did not guarantee high adoption. Zhou et al. (2017) studied 134 counties in the United States and tested 18 factors against 2014 plug-in vehicle market shares. Income, extreme temperature, incentives and carpool lane access mattered for almost every price group, but other factors were different for each group. Narassimhan and Johnson (2018) used 2008 to 2016 purchase data from U.S. states. They found that tax incentives and public charging both raised plug-in purchases per person, that rebates worked better than tax credits, and that BEV and PHEV purchases reacted differently. Lane et al. (2018) surveyed drivers who planned to buy or lease a car. They found that people interested in PHEVs cared mostly about saving money on fuel and maintenance, while people interested in BEVs cared more about the environment and new technology. The study closest to this project is Gifford and Barbier (2025). They used 2024 Washington State registrations by census tract and found that longer commutes were linked to fewer BEVs, while more public chargers, higher income and more education were linked to more EVs.

These studies mostly explain how much EV adoption happens and which policy, charging and income factors are linked to it. They measure adoption as purchases per person or market share, and they group the data by country, state, county or census tract. I did not find a study that describes what the registered EV fleet in a state actually looks like: which makes and models dominate, how BEVs and PHEVs split, and how electric range varies across models. Even the Washington study focuses on commuting and chargers, not on makes, models or range. This project fills that gap. It describes the registered EV fleet in Washington State by make, model,

EV type, electric range, model year and county, using the registration data from the Washington State Department of Licensing.

2. Analysis and Modeling Methods

Data source and content

The data is the Electric Vehicle Population Data from the Washington State Department of Licensing (DOL). It lists the BEVs and PHEVs that are currently registered in Washington State. The raw file had 300,142 records and 16 fields. This project uses county, city, model year, make, model, EV type and electric range, plus one field I created called range status.

Data cleaning

I loaded the raw file into PostgreSQL and cleaned it with SQL. Table 1 shows each problem I found and what I did about it.

Table 1. *Data cleaning log for the Electric Vehicle Population Data (raw N = 300,142)*

Issue	Records	What I did	Why
Registered outside Washington	789	Removed	Not part of this study. These records also explain all 15 records with a missing county, city and postal code, and 789 of the 790 missing legislative districts.
Electric range = 0 (all BEVs)	196,594	Set to blank and labeled "Not researched"; kept	A 0 means the DOL has not researched the range. It does not mean the vehicle has no range. The dataset gives the same records the label "Eligibility unknown as battery range has not been researched."
Electric range blank	35	Labeled "Not researched"; kept	Same reason as above.
Legislative district missing (Washington)	1	Kept	This field is not used.
Duplicate records	0	Nothing to do	Checked full rows and DOL Vehicle ID.
Final analysis dataset	299,353		

Vehicles with no researched range stay in every count by make, model, EV type, county and model year. They are left out only from the electric range statistics.

Analysis methods

The research question asks how EVs are spread across models, makes, range and location, so this project uses descriptive and exploratory analysis. I group the registrations by make, model, EV type, county and model year, and I calculate counts and percentages. For electric range, I calculate the mean, median, minimum, maximum and standard deviation for BEVs and PHEVs separately, using only the 102,724 vehicles that have a researched range. Range values are skewed and very different for BEVs and PHEVs, so I report the median next to the mean and I never mix the two EV types into one average. [EDIT THIS SENTENCE TO MATCH YOUR PLAN: For the final report, I plan to compare range across makes and model years using regression or analysis of variance, and to compare counties after adjusting for population.]

Software tools

I used three tools. PostgreSQL 18 (through pgAdmin) loaded and cleaned the data and produced the summary tables. Tableau built the charts and the county maps. Microsoft Excel was used to export the dataset to a CSV file for loading into PostgreSQL.

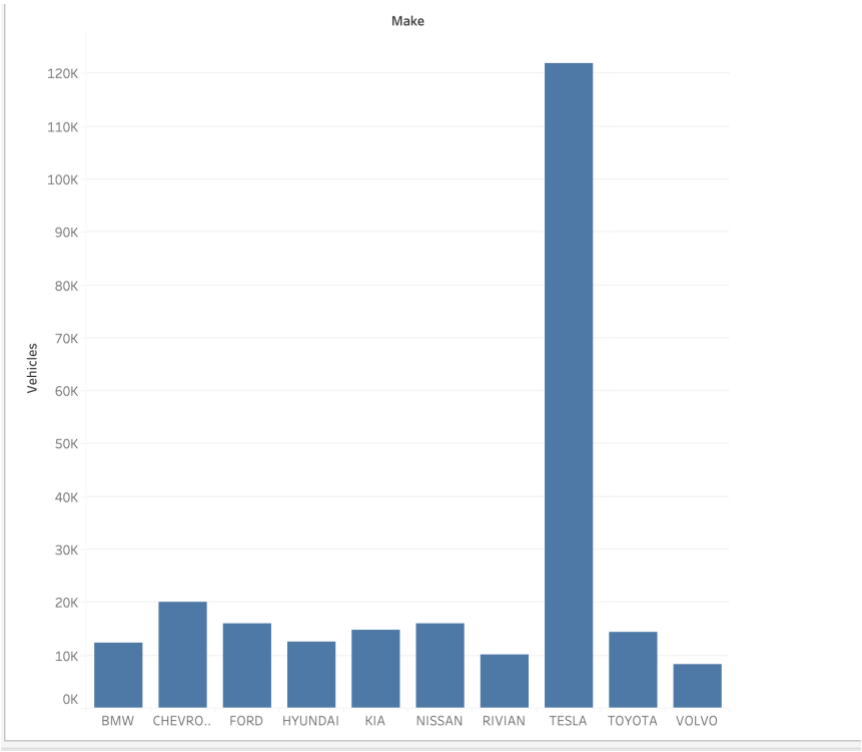
3. Descriptive Statistics and Visualizations

After cleaning, the dataset has 299,353 registered EVs from 52 makes and 205 models in 39 counties. Battery electric vehicles make up 241,238 vehicles (80.6%) and plug-in hybrids make up 58,115 (19.4%) (Figure 2).

Makes and models

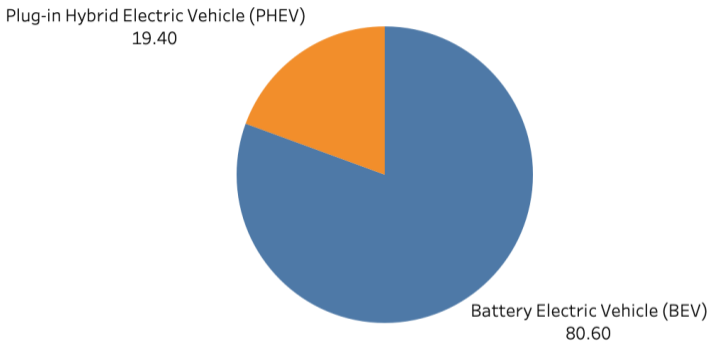
Tesla has 121,879 vehicles, which is 40.7% of all registered EVs and about six times more than Chevrolet, the second-largest make with 20,177 vehicles (6.7%). The top 10 makes together make up 82.3% of all registrations (Figure 1). The two most common models are both Teslas: the Model Y (66,379 vehicles, 22.2%) and the Model 3 (38,735 vehicles, 12.9%). Together they make up 35.1% of the fleet (Table 2).

Figure 1. *Top 10 EV makes in Washington State (N = 299,353 registered vehicles)*



Note. Tesla alone has 121,879 vehicles; every other make has fewer than 21,000.

Figure 2. *EV type distribution in Washington State (N = 299,353 registered vehicles; values are percentages)*



Note. BEV = battery electric vehicle (241,238 vehicles); PHEV = plug-in hybrid electric vehicle (58,115 vehicles).

Table 2. *Top 10 EV models in Washington State (N = 299,353)*

Rank	Make and model	Vehicles	Share of all EVs
1	Tesla Model Y	66,379	22.2%
2	Tesla Model 3	38,735	12.9%
3	Nissan Leaf	13,357	4.5%
4	Hyundai Ioniq 5	7,734	2.6%
5	Tesla Model S	7,730	2.6%
6	Chevrolet Bolt EV	7,585	2.5%
7	Tesla Model X	7,018	2.3%
8	Ford Mustang Mach-E	6,827	2.3%
9	Volkswagen ID.4	6,343	2.1%
10	Toyota RAV4 Prime	5,698	1.9%

Electric range

Only 102,724 vehicles (34.3% of the fleet) have a researched range. That includes 58,080 of the 58,115 PHEVs but only 44,644 of the 241,238 BEVs (18.5%), and the researched BEVs are mostly older model years. Table 3 shows the results. BEVs have a median range of 215 miles and PHEVs have a median of 32 miles. Among researched vehicles, Tesla has the highest average range of the larger makes (241.5 miles, n = 23,659), followed by Chevrolet (152.6 miles, n = 9,793).

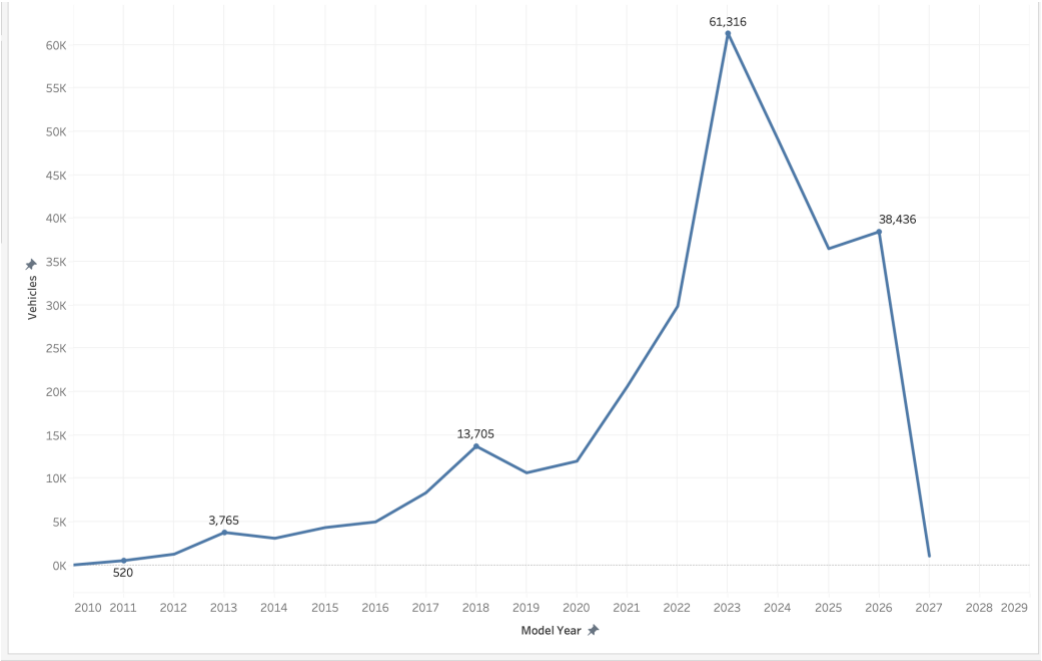
Table 3. *Electric range (miles) for vehicles with a researched range, by EV type (n = 102,724)*

EV type	n	Mean	Median	Min	Max	Std. dev.
BEV	44,644	200.3	215.0	29	337	71.9
PHEV	58,080	32.3	32.0	1	153	14.0

Model year

Registrations grew from 520 vehicles of model year 2011 to a peak of 61,316 vehicles of model year 2023 (Figure 3). The lower counts for model years 2024 to 2027 probably mean those model years are still being registered. They do not show a real drop in EV adoption.

Figure 3. *EV registrations by model year in Washington State, 2010 to 2027 (N = 299,323 registered vehicles)*

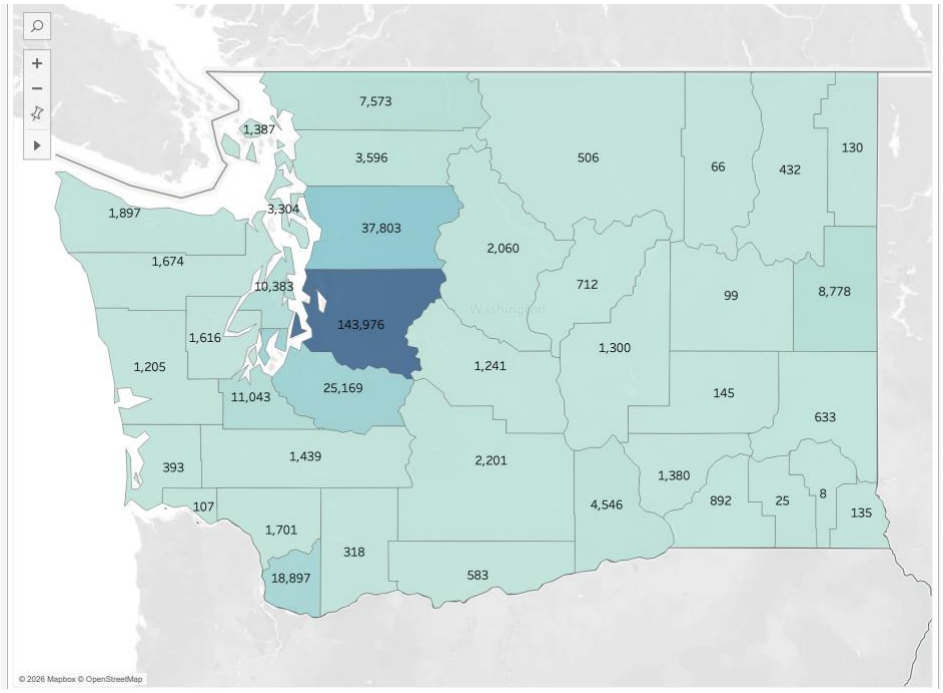


Note. Excludes 30 vehicles with model years before 2010. Model year 2027 has only 1,040 vehicles because those models are just starting to be registered.

Location

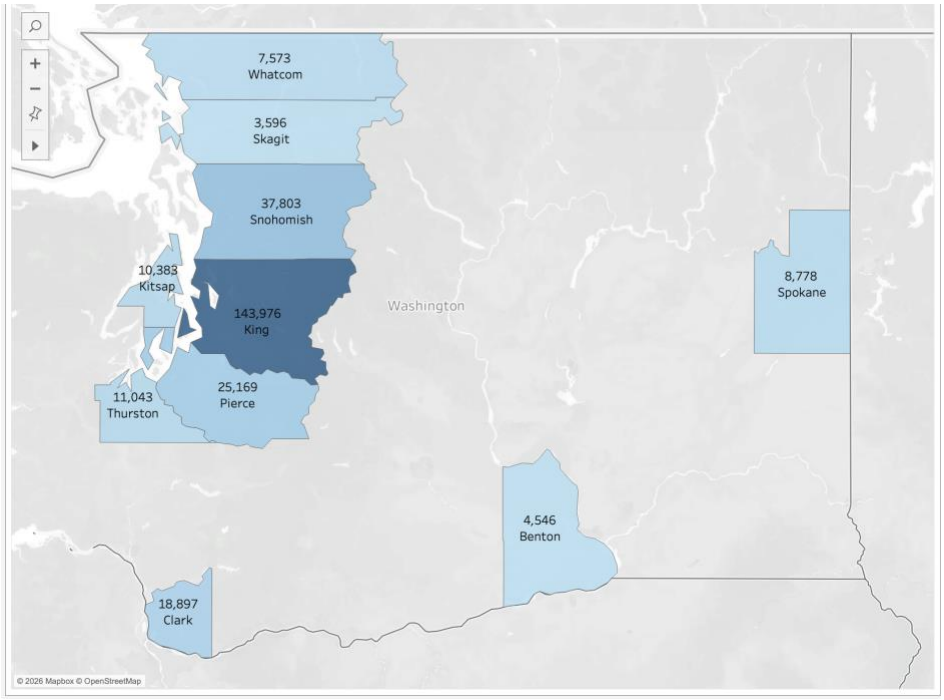
EV registrations are heavily concentrated in the Puget Sound region (Figure 4). King County alone has 143,976 vehicles (48.1%). King, Snohomish (37,803 vehicles, 12.6%) and Pierce (25,169 vehicles, 8.4%) counties together hold 69.1% of all EVs. The ten largest counties hold 90.8% (Figure 5). Spokane County, with 8,778 vehicles (2.9%), is the largest county in eastern Washington. These are raw counts and not rates per person, so larger counties naturally have more vehicles.

Figure 4. *EV registrations by county in Washington State (N = 299,353 registered vehicles)*



Note. Darker shading means more registered vehicles. The number in each county is its vehicle count.

Figure 5. *Top 10 counties by EV registrations in Washington State (n = 271,764 of 299,353 registered vehicles)*



Note. The other 29 counties are not shown.

Limitations

The DOL data is a snapshot of vehicles that are registered right now. It does not include vehicles that were sold out of state or scrapped. Range is missing for most BEVs, so the BEV range results mainly describe older vehicles.

References

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Sierzchula, W., Bakker, S., Maat, K., & van Wee, B. (2014). The influence of financial incentives and other socio-economic factors on electric vehicle adoption. *Energy Policy, 68*, 183–194. https://doi.org/10.1016/j.enpol.2014.01.043

Washington State Department of Licensing. (n.d.). *Electric vehicle population data* [Data set]. Data.WA.gov. [ADD THE URL FROM THE PAGE YOU DOWNLOADED THE DATA FROM]

Zhou, Y., Santini, D. J., Vazquez, K., & Rood, M. (2017, January 8–12). *Contributing factors in plug-in electric vehicle adoption in the United States: A metro/county level investigation* [Paper presentation]. Transportation Research Board 96th Annual Meeting, Washington, DC, United States.